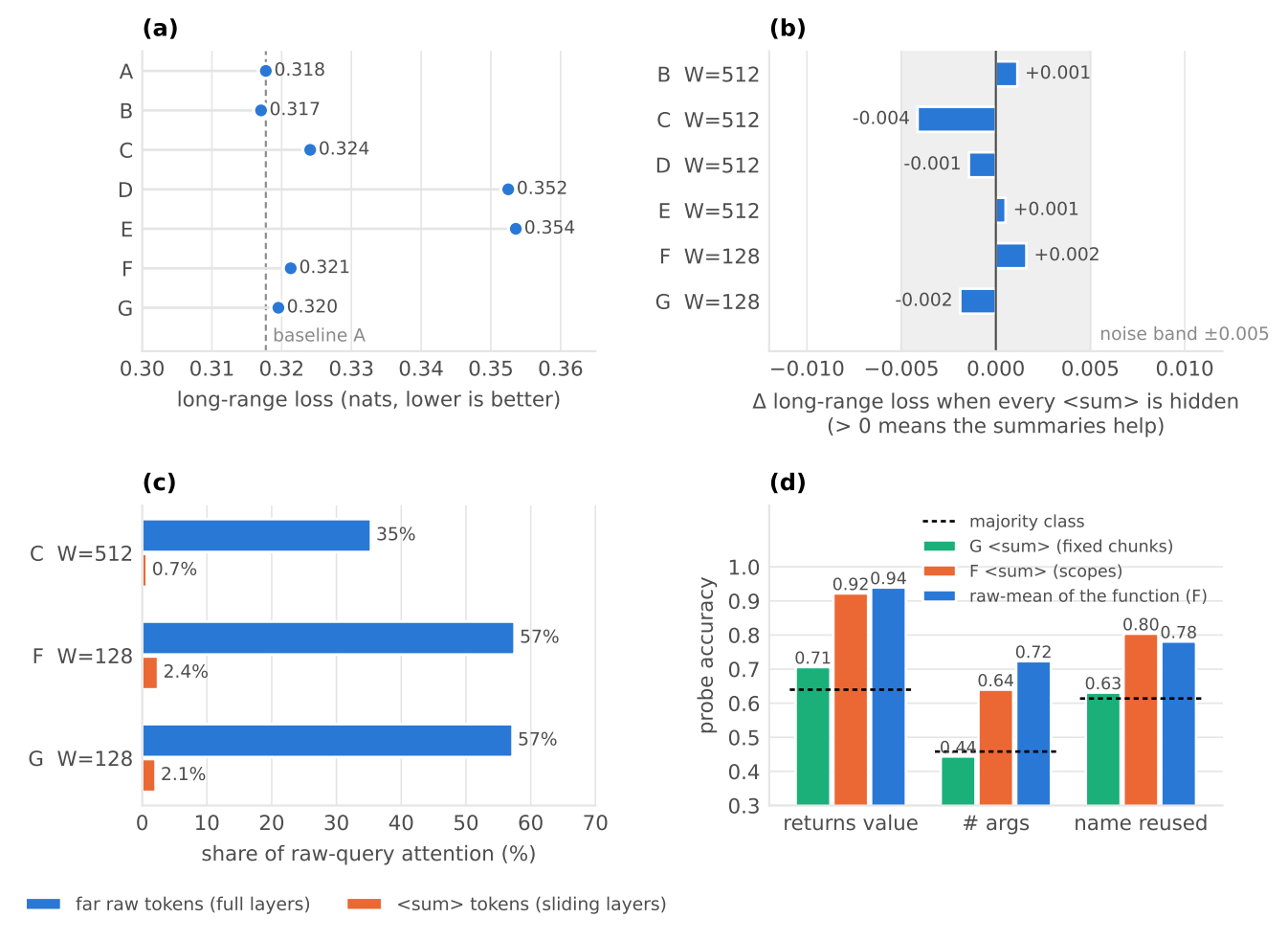
(a)
A
B
C
D
E
F
G
0.318
0.317
0.324
0.352
0.354
0.321
0.320
baseline A
0.30
0.31
0.32
0.33
0.34
0.35
0.36
long-range loss (nats, lower is better)
(b)
B W=512
C W=512
D W=512
E W=512
F W=128
G W=128
+0.001
-0.004
-0.001
+0.001
+0.002
-0.002
noise band ±0.005
−0.010
−0.005
0.000
0.005
0.010
Δ long-range loss when every <sum> is hidden
(> 0 means the summaries help)
(c)
C W=512
F W=128
G W=128
35%
0.7%
57%
2.4%
57%
2.1%
0
10
20
30
40
50
60
70
share of raw-query attention (%)
far raw tokens (full layers)
<sum> tokens (sliding layers)
(d)
majority class
G <sum> (fixed chunks)
F <sum> (scopes)
raw-mean of the function (F)
probe accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.71
0.92
0.94
0.44
0.64
0.72
0.63
0.80
0.78
returns value
args
name reused